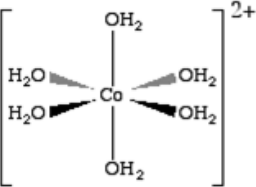
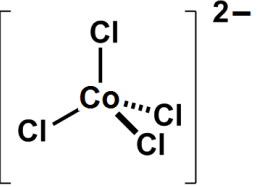


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9.	(a)			 (1)  (1)						
	(b)			ligands cause d-orbitals to split into three lower and two higher energy levels (1) electrons absorb energy and are promoted to a higher energy level (1) colour seen is the colours not absorbed (1)	2			2		
					3			3		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)			white precipitate (1) silver chloride forms / silver ions cause a precipitate with chloride (1) solution turns pink (1) removal of chloride causes equilibrium to shift to left (1)		1 1	 1 1	 4		1 1
	(d)	(i)		$f = 5.79 \times 10^{14}$ (1) $E = hf = 3.84 \times 10^{-19}$ (1) $E = 231$ (1)		3		3	3	
		(ii)		value above 600 nm as this is absorbed by chloro but not aqua / value below 560 nm as this is absorbed by aqua complex but not chloro		1		1		1

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)		$K_c = \frac{[[CoCl_4]^{2-}] \times [H_2O]^6}{[[Co(H_2O)_6]^{2+}] \times [Cl^-]^4} \quad (1)$ $mol^2 dm^{-6} \quad (1)$		2		2	1	
		(iv)		concentration of aqua complex = 0.596 concentration of chloride = 0.224 / concentration of chloro complex = 0.124 award (2) for all three concentrations correct award (1) for any two correct $K_c = 14.0 \quad (1)$			3	3	3	
				Question 9 total	5	8	5	18	7	3